

Hadoop vs Conventional HPC (MPI + PBS): A Comparative Analysis of High-Performance Data Processing Systems

Evelyn Goh Yuan Qi
Faculty of Computing
University of Technology Malaysia
evelynyuan@graduate.utm.my

Abstract— This paper gives a comparative analysis of the conventional High-Performance Computing (HPC) systems in terms of MPI and PBS and the Hadoop distributed computing framework. The paper explores their architectural differences, their performance features, scalability, fault tolerance and their ability to accommodate various workloads. Conventional HPC systems are very proficient with compute-intensive tasks, as they are tightly coupled and optimized to handle parallel processing, unlike Hadoop, which is geared towards data-intensive tasks, through distributed storage and processing via HDFS and MapReduce. The results have shown that Hadoop enhances scalability and fault tolerance, but at the cost of introducing performance overhead to computationally intensive jobs. This paper also points out that the two methods are complementary and their combination can offer more effective solutions to the contemporary data processing landscape, especially in the era of big data and artificial intelligence.

Keywords— *High-Performance Computing (HPC), Hadoop, MPI, Distributed Computing, Big Data Processing*

I. INTRODUCTION

In today's digital era, the volume of data output created by applications like social media, scientific simulations, and enterprise systems is growing rapidly, and thus the need to use effective solutions to process the data has become a major requirement. The conventional computing systems are usually ineffective to manage the sheer amount, speed and diversity of the current data and this has given rise to High-Performance Computing (HPC) as a key strategy towards addressing problems that are computationally intense in nature. The real-world industries, including the automotive sector, where big volumes of sensor and IoT data need scalable platforms like Hadoop, are becoming the new drivers of big data applications (Luckow et al., 2015). The HPC systems find extensive applications in science and engineering, where parallel computing methods and architectures, including clusters, shared-memory, and GPUs, are used to obtain high computational performance (Ivanović et al., 2016).

Conventional HPC systems generally use parallel programming models like Message Passing Interface (MPI) and job scheduling software like Portable Batch System (PBS). These systems are intended to perform complex numerical simulations effectively, especially in structured and compute-intensive environments. Nevertheless, as the concept of big data gains more and more popularity, the conventional HPC architecture is limited in its ability to process large-scale unstructured data and guarantee scalability and fault tolerance. Moreover, HPC systems typically prioritize performance, which may limit flexibility in being able to adapt to current data-driven applications (Peisert, 2017).

To overcome these challenges, distributed computing platforms like Hadoop have been presented. Hadoop is an architecture that is designed to handle large scale data processing and storage relying on distributed file systems and parallel processing frameworks like MapReduce. It runs on groups of commodity hardware and offers functionality, including scalability, fault tolerance, and cost-effectiveness, and is applicable to big data analytics and large-scale data processing tasks (Xie et al., 2018). Moreover, the architecture of Hadoop, mainly the Hadoop Distributed File System (HDFS), facilitates the credible storage of data by replicating and distributing data among several nodes (Sivaraman and Manickachezian, 2014).

Since the conventional HPC systems and the modern distributed systems differ, one should be aware of the strengths and weaknesses of each of them in the high-performance data processing environments. With more organizations becoming dependent on data-driven decision-making, choosing a suitable processing framework is now crucial to achieve performance efficiency, scalability, and cost-effectiveness. Therefore, this paper aims to compare Hadoop with conventional HPC systems based on MPI and PBS, focusing on their performance characteristics, scalability, fault tolerance, and suitability for different types of workloads. Through this comparative analysis, the study seeks to highlight how Hadoop resolves the limitations of conventional HPC through distributed storage and parallel processing mechanisms. In addition, this paper discusses different scenarios in which each approach achieves optimal performance.

II. MAIN BODY

A. Conventional High-Performance Computing

Conventional High-Performance Computing (HPC) systems are created to tackle the intricate computational issues through parallel processing involving a number of nodes. They are usually employed in scientific simulations and engineering-related applications, where the computational power is in high demand. HPC systems are usually made up of clustered computing resources linked to the high-speed network to facilitate efficient large-scale processing (Luckow et al., 2016).

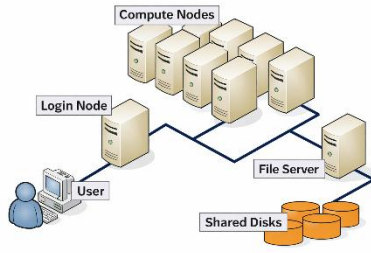


Fig. 1. Conventional High-Performance Computing (HPC) Architecture (adapted from Luckow et al., 2016)

Figure 1 shows an example of a conventional HPC architecture, with jobs being posted via a login node and processed by a number of compute nodes linked together by a fast network and with common storage to assist with data access.

One of the fundamental elements of HPC is the Message Passing Interface (MPI), which allows communication among processes by exchanging messages. Although MPI can deliver high performance in programs with a high rate of computation, the low-level communication and synchronization must be handled by programmers, making it more complex to develop (Yu et al., 2013). Moreover, job scheduling systems like Portable Batch System (PBS) are employed to coordinate resource allocation and execute jobs in batch mode which can be associated with delays because of job queuing (Peisert, 2017).

Despite their efficiency, conventional HPC systems have limitations in dealing with modern data-intensive workloads. They also have low tolerance to fault and are not as flexible as newer distributed computing frameworks at handling large, unstructured data.

B. Hadoop Framework

Hadoop is a distributed computing platform that is used to store and process big data on groups of commodity hardware. Hadoop is data-intensive workload optimized, unlike conventional HPC systems, and enables scalable and cost-effective data processing. Its architecture is based on the Hadoop Distributed File System (HDFS) for storage and MapReduce programming model to process parallel data (Xie et al., 2018).

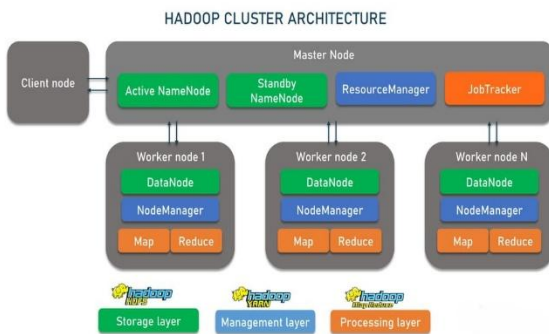


Fig. 2. Hadoop Cluster Architecture (adapted from Xie et al., 2018)

Figure 2 illustrates the Hadoop architecture, involves a master node that coordinates, and data is distributed to several DataNodes. The MapReduce model is used to process the data, which allows distributed and parallel processing.

HDFS makes use of data replication to ensure fault tolerance, it splits the data into blocks and then distributes the

data blocks into various nodes. This makes certain that the information is available even in the event of failure of individual nodes. The MapReduce model operates in two phases: the map phase where input data is processed into intermediate key-value pairs, and the reduce phase where the results are aggregated to give final outputs (Sivaraman & Manickachezian, 2014).

Moreover, Hadoop has a resource management layer, such as YARN, which provides system resources and coordinates job execution on the cluster. This enables Hadoop to effectively work with large scale, unstructured and heterogeneous data. Nevertheless, Hadoop can be limited in performance over iterative and real-time tasks when compared to the conventional HPC system due to its scalability and fault tolerance characteristics.

C. Comparative Analysis

A comparison between conventional HPC systems and Hadoop highlights some major differences in the design, performance, and the applicability to various workloads as demonstrated in Table 1. Conventional HPC systems are best suited to compute-intensive programs with the help of parallel processing model like MPI, while Hadoop is tailored to data-intensive programs with the help of a distributed computing model. These variations affect their scalability, cost, fault tolerance and efficiency in general.

TABLE I. COMPARISON BETWEEN TRADITIONAL HPC AND HADOOP

Criteria	Traditional HPC	Hadoop
Processing Model	Parallel computing (MPI)	Distributed computing (MapReduce)
Scalability	Limited	Highly scalable
Cost	Expensive hardware	Commodity hardware (low cost)
Fault Tolerance	Low (checkpointing needed)	High (data replication in HDFS)
Data Type	Structured data	Structured & unstructured data
Performance	Very fast for computation	Slower for computation-heavy tasks
Flexibility	Low	High

Conventional HPC systems are more performance efficient in compute intensive tasks because they have tightly coupled designs and are highly efficient in parallel processing with MPI. Nevertheless, they are not flexible and scalable to process large amounts of heterogeneous data. Conversely, Hadoop is more appropriate in processing big data since it is highly scaled and fault tolerant because it disperses data and computation among more than one node. However, Hadoop can also have a performance overhead caused by disk-based processing, and is not as efficient as needed by tasks that need heavy computation.

Although Hadoop has improved numerous limitations of the conventional HPC regarding big data, it does not entirely displace HPC systems. Rather, the two methods have their purposes. HPC is still crucial in scientific simulations and high-performance numerical calculations, whereas Hadoop is better in data analytics and large-scale data processing. Recent studies even propose the combination of the two systems to take advantage of their synergistic capabilities and process efficiently both compute-intensive and data-intensive tasks (Luckow et al., 2016).

D. Real-World Case Study: Hadoop in the Automotive Industry

According to Luckow et al. (2015), Hadoop has found extensive applications in real world scenarios with the automotive sector being one of the most popular. For example, BMW deploys data platforms based on Hadoop to process and analyze massive amounts of data created by connected vehicles, sensors, and Internet of Things (IoT) devices. The modern vehicles are capable of producing huge quantities of data that relates to driving behavior, vehicle diagnostics and the conditions of the environment, and need scalable storage and processing systems.

Through the distributed architecture of Hadoop, BMW can easily store and process terabytes to petabytes of data to use in predictive maintenance, traffic analytics and automated driving assistance. Hadoop allows combining heterogeneous data sources and provides high-quality analytics, which will allow the company to ensure operational efficiency and improve the quality of decision-making. This illustrates the fact that Hadoop is better than conventional HPC systems in dealing with large-scale and data intensive workloads in real world industry settings.

E. Real-World Case Study: Conventional HPC in Multi-Scale Simulation Systems

The traditional HPC systems are extensively used in the real-life scientific computing devices, especially in the multi-scale simulation scenarios. For example, Ivanović et al. (2016) explain the application of HPC systems in multi-scale muscle modeling, involving complex biological processes that are simulated with the help of parallel computing. These systems make use of Message Passing Interface and GPU acceleration to share the computational loads to several processors, which is useful in efficiently solving extremely complicated numerical models.

This system uses macro level and micro level models to conduct large scale simulations, which demand high computational power and parallel processing ability. HPC can enable the researcher to have high scalability and performance with significant time savings in processing time. This proves that the conventional HPC systems are very efficient in applications that are compute intensive and demand high precision calculations and high processing rate. These systems are, however, not as applicable to large-scale, unstructured data as they are limited to compared to distributed systems such as Hadoop.

III. CONCLUSION

In conclusion, this paper has compared the use of conventional High-Performance Computing (HPC) systems (MPI and PBS) with Hadoop as a contemporary distributed computing platform. The analysis indicates that traditional HPC systems are very efficient with tasks that are compute

intensive specifically in scientific and engineering applications, as the systems are optimized in terms of parallel processing. Nevertheless, they are limited in scalability, flexibility, and fault tolerance to large-scale, data-intensive workloads.

Conversely, Hadoop offers an inexpensive, fault-tolerant and scalable solution to handling large amounts of structured and unstructured data. Its distributed nature and data replication capabilities render it optimal to big data analytics. Nonetheless, Hadoop is unlikely to be as high-performing as HPC systems in cases of computation-intensive tasks.

From this comparative analysis, it can be observed that both approaches are not competing replacements but complementary technologies. The future computer systems would probably combine the best features of HPC and Hadoop in the present age of artificial intelligence and data-driven applications. The future of conventional HPC systems will probably involve adding more data-friendly features and enhanced resilience, whereas Hadoop and its ecosystem might become more in-memory processing, real-time analytics, and tighter integration with AI frameworks.

Moreover, additional architectural components like hybrid computing designs, smart resource management, and AI-based optimization methods can be implemented to optimize system performance and efficiency. Such advances imply that high-performance data processing environments of the future will not be based on one method, but instead it will be based on combining various technologies to address the increasing needs of both compute-intensive and data-intensive workloads.

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Logbook

Name: Evelyn Goh Yuan Qi

Student ID: A23CS0222

Date	Search Keywords	Database / Tools Used	AI Prompts Used	Task Description
6 April 2026	High Performance Computing architecture MPI cluster computing	Google Scholar, IEEE Xplore, Scopus	Explain what High Performance Computing (HPC) is	Conducted initial research on High Performance Computing concepts, including cluster computing, MPI, and parallel processing, to understand the topic area.
8 April 2026	Hadoop vs HPC comparison big data processing	Google Scholar, IEEE Xplore, Scopus	What is the difference between Hadoop and HPC	Identified Hadoop as a distributed computing framework suitable for big data processing. Selected the topic comparing Hadoop with traditional HPC systems.
9 April 2026	Hadoop architecture HDFS MapReduce YARN	IEEE Xplore, Scopus	Explain Hadoop architecture (HDFS + MapReduce)	Reviewed academic papers on Hadoop architecture and key components including HDFS and MapReduce.
10 April 2026	HPC MPI PBS job scheduling limitations	Google Scholar, Scopus	Explain MPI and PBS in simple terms	Studied traditional HPC systems focusing on MPI programming model and PBS job scheduling mechanisms.
11 April 2026	Hadoop vs HPC performance comparison	IEEE Xplore	Compare Hadoop and HPC differences	Analyzed key differences between Hadoop and HPC systems and started drafting ideas for comparison.
12 April 2026	Hadoop real world application automotive big data	Scopus	Example of Hadoop real world use	Identified a real-world case study (BMW automotive data) and prepared content for integration into the report.
13 April 2026	academic writing introduction HPC big data	Google Scholar, CHATGPT	Help improve introduction paragraph	Drafted the introduction section including background, problem statement, and research objective.
14 April 2026	HPC MPI PBS limitations parallel computing	IEEE Xplore, Google Scholar, CHATGPT	Explain MPI and PBS in simple terms	Wrote the Traditional HPC section, explaining architecture, MPI communication model, PBS scheduling, and system limitations.
15 April 2026	Hadoop architecture HDFS	IEEE Xplore	Explain Hadoop	Completed the Hadoop framework section, including HDFS, MapReduce, YARN,

	scalability fault tolerance		architecture clearly	and key features such as scalability and fault tolerance.
16 April 2026	HPC vs Hadoop comparison criteria table	CHATGPT	Generate comparison table structure	Developed the comparison table and comparative analysis based on performance, scalability, cost, and flexibility.
17 April 2026	academic conclusion Hadoop vs HPC	CHATGPT	Help write conclusion	Completed the conclusion summarizing key findings and differences between HPC and Hadoop.
18 April 2026	APA citation format journal IEEE	Google Scholar, CHATGPT	Format references APA 7th edition	Formatted all references in APA style and verified consistency between in-text citations and reference list.
19 April 2026	proofreading grammar academic writing	None	Improve academic tone	Conducted final proofreading, grammar correction, and formatting according to IEEE template before submission.